Ashton Harrell

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Education

Louisiana Tech University — Ruston, LA

Bachelor of Science in Computer Science, Minor in Mathematics — *Aug. 2023 – Nov. 2026*

- **Relevant Coursework:** Artificial Intelligence, Software Engineering, Senior Capstone, Database Management Systems, Programming Languages, Intro. to Cybersecurity, Advanced Data Structures & Algorithms, Intro. to Graph Theory
- **Honors and Awards:** 3rd Place in LwC Freshman Exhibition 2024; Recipient of TC Energy STEM Scholarship

Experience

Resident Assistant — June 2023 – Present

Louisiana Tech University — *Ruston, LA*

- Developed and maintained residential communities across campus housing and apartment facilities
- Planned, organized, and executed department- and building-scale events with up to 100+ residents
- Enforced university policies among residents and staff; administered prospective student housing tours

Lead Sales Associate — May 2023 – Aug. 2023

Dollar General — *Epps, LA*

- Administered POS transactions with customers and distributors
- Managed facility operations in the absence of a supervisor
- Supervised and trained new employees while ensuring assigned work was completed

Projects

Catalyst | *TypeScript, Next.js, React, Firebase, Google GenAI* — Summer 2026 - Present

- Developed a full-stack AI study companion that centralizes course materials, deadlines, notes, and syllabi for college students
- Built AI-powered tools to generate quizzes, flashcards, personalized study plans, and contextual academic support from uploaded materials
- Implemented OCR and document-processing workflows, calendar and email reminders, two-factor authentication, analytics dashboards, Docker deployment, and automated testing

Transformer Recreation | *Python, PyTorch, NLP* — Summer 2026

- Implemented a Transformer language model from scratch in PyTorch, including scaled dot-product attention, multi-head attention, positional encoding, and encoder–decoder components
- Built an end-to-end NLP training pipeline that downloads and preprocesses IWSLT'14 English–German data, trains a BPE tokenizer, and creates batched token sequences
- Developed autoregressive GPT-style text generation with checkpointing, temperature sampling, and top-k decoding to evaluate language-generation capabilities

Zono | *TypeScript, React Native, Expo, Supabase, Ollama* — Spring 2026

- Collaborated on a cross-platform mobile application for real-time location sharing, messaging, and interactive activities
- Developed Android and iOS features using React Native and Expo, with Supabase supporting application data and backend services
- Integrated an optional locally hosted AI chatbot through Ollama and configured environment-based API credentials

Technical Skills

Languages: Java, Python, C/C++, TypeScript, SQL, HTML/CSS, Rust
Frameworks & Libraries: Next.js, React, Expo, Node.js, Flask, PyTorch
Databases: Firebase, Supabase, PostgreSQL, NoSQL, MySQL
Developer Tools: Git, GitHub, VS Code, Docker, NeoVim (Vim), Ollama
Concepts: Neural Networks, Machine Learning, OOP, REST APIs, Agile/Scrum, RAG, NLP